MILITARY SPECIFICATION

TRUCKS, LIFT, FORK, DIESEL-ENGINE-DRIVEN:

PNEUMATIC-TIRED, ROUGH TERRAIN; 10,000 POUND CAPACITY

AT 48-INCH LOAD CENTER

This amendment forms a part of MIL-T-52843C, dated 1 May 1985, and is approved for use by all Departments and Agencies of the Department of Defense.

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3.9.2, delete in its entirety and substitute the following:

3.9.2 Vehicle floatation. The truck shall have a Floatation Index (FI) of not more than 25 with rated load in carry position and without rated load. The FI shall be calculated using the following formula:

$$FI = \text{Antilog} \left[0.211 - 0.350\,C + 0.00710\,P - \frac{0.0752(P_F - P_R)}{P} - \frac{0.0402(A_F - A_R)}{A} + 0.0346K \right]$$

Where:

P_F = Tire pressure, front (psi)
P_R = Tire pressure, rear (psi)
P = Tire pressure, average = 1/2 (P_F + P_R)
A_F = Axle load, front (lbs)
A_R = Axle load, rear (lbs)
A = GRoss vehicle weight - A_F + A_R
K = 53.75 + 0.000139A (for nonradial tires)
K = 55.77 + 0.000032A (for radial tires)

$$C + \text{Log}\left[\frac{A}{0.607P + 1.35\left(\frac{117R}{5W + D}\right) - 4.93} \right]$$

AMSC N/A FSC 3930

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R = Tire ply rating
W = Tire width (in. – as cited in tire size)
D = Rim diameter (in. – as cited in tire size)

The FI shall be determined with forklift fully fueled and serviced with lubricants. For the purpose of calculating both loaded and unloaded FI, the front tire inflation pressures shall be based on front axle weight with rated capacity loaded in carry position and rear tire inflation pressures shall be based on rear axle weight with no load on forks in carry position. Pressure may be reduced by not more than 30 percent of the recommended pressure published in the TRA Yearbook for the forklift's tire load at 5 mph rating.

Custodians:
Army - ME
Navy - SA
Air Force - 99

Preparing activity:
Army - ME

Project 3930-0628

Review activities:
Army – AT, MT
Air Force – 84
DLA - CS

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